Grapevine leafroll: roguing, testing and replanting

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1 Overview

Grapevine leafroll disease is caused by viruses that mealybugs carry from vine to vine along the trellis. An infected vine yields less and worse fruit for the rest of its life. Infection stays invisible for a year or more, so roguing symptomatic vines may come too late. Atallah et al. (2015) built a vine-level cellular automaton of a vineyard with an economic layer. They compared eighteen strategies of roguing, replanting and virus-testing by their expected net present value (ENPV) over 50 years. Their finding is that testing the neighbours of each symptomatic vine pays, by about 18%, while roguing alone does not.

The journal article is behind a paywall, so this model follows Chapter 2 of the first author's dissertation (Atallah 2014), which reports the same design and results. It is an adaptation. The sources leave unstated when a symptomatic vine is found and what state the first infections start in, and those choices move the results.

```
ENV["GKSwstype"] = "100"
using EcologicalCellularAutomata
using CellularAutomataCore
using Plots
using Statistics
```

2 The model

A 5.2-acre vineyard has 44 trellis rows of 130 vines. Each vine is healthy, exposed but undetectable, exposed and detectable, infective-moderate or infective-high, and has an age in months. Each month:

- a healthy vine is infected at rate 4.2 per infective neighbour along its trellis row and 0.014 per infective neighbour in the next rows;
- an infected vine stays undetectable for 4 to 18 months (triangular, mode 12);
- an exposed-detectable vine becomes infective-moderate at a mean of 24, 48 or 72 months for young, mature or old vines;
- infective-moderate becomes infective-high at a mean of 36 months.

A vine earns from 36 months of age: $0.43 a month if healthy, less if infected (30%, 50% or 75% less yield and 10% less in price). Roguing and replanting costs $7.25 and a virus test $2.61, discounted at 0.9959 a month.

```
model = build(:grapevine_leafroll)
```

```
CAModel grapevine_leafroll
  Atallah et al. (2015) grapevine leafroll, undetectable period as duration
  states:
    state::Int8 = 0 ∈ 0:4  # 0 H, 1 exposed-undetectable, 2 exposed-detectable, 3
    infective-moderate, 4 infective-high
    age::Int16 = 0  # age (months)
    clock::Int16 = 0  # months left undetectable, or months infective
    rogued::Bool = false  # rogued and replanted this month
    tested::Bool = false  # virus-tested this month
  parameters:
    α = 4.2  # infection rate per infective neighbour along the trellis row
    (1/month)
    β = 0.014  # infection rate per infective neighbour across rows (1/month)
    eu_min = 4.0  # shortest undetectable period (months)
    eu_mode = 12.0  # commonest undetectable period (months)
    eu_max = 18.0  # longest undetectable period (months)
    L_young = 24.0  # mean latency of young vines (months)
    L_mature = 48.0  # mean latency of mature vines (months)
    L_old = 72.0  # mean latency of old vines (months)
    inf_period = 36.0  # mean time infective-moderate (months)
  rule: LocalRule(<anonymous function>, Offsets((-1, 0), (1, 0), (0, -1), (0, 1)))
```

3 The economics without disease

Without disease the vineyard's value is fixed. It is the check on the age and time conventions:

```
nodisease = grapevine_npv(solve(problem(:grapevine_leafroll; infected=0.0)))
round(nodisease / 1000; digits=1)
```

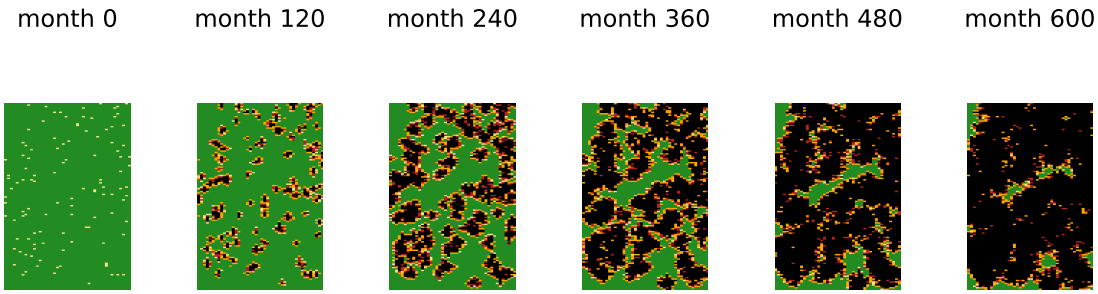
```
462.7
```

That is the source's $463k.

4 Spread without control

Two per cent of the vines start infected. Transmission along a trellis row is almost certain within a month of a neighbour becoming infective, and across rows it is rare. But a newly infected vine takes about three years, on average, to become infective itself. That slows the spread along a row to a few vines a decade, and over the same decades the rare jumps across rows add up, so the patches grow in both directions:

```
sol = solve(problem(:grapevine_leafroll; seed=2); saveat=120)
vinemap(u, title) = heatmap(u.state; c=[:forestgreen, :khaki, :orange, :firebrick,
:black],
                            clims=(0, 4), colorbar=false, framestyle=:none, title,
                            aspect_ratio=0.5)
plot([vinemap(u, "month $t") for (t, u) in zip(sol.t, sol.u)]...; layout=(1, 6),
     size=(1000, 330))
```



Each column is a trellis row. Healthy vines are green, exposed vines yellow (undetectable) or orange (detectable), and infective vines red (moderate) or black (high).

The vineyard's half-life is the time until half the vines are infected. The source reports 188 months without control:

```
halflives = [grapevine_half_life(solve(problem(:grapevine_leafroll; seed=s))) for s
in 1:20]
(mean=mean(halflives), sd=round(std(halflives); digits=1))
```

```
(mean = 235.2, sd = 6.2)
```

The disease is slower here, by about four years.

5 Strategies

The source's Table 2.5 gives the ENPV of each strategy over 1000 runs. Twenty runs each here, for the main ones:

- `:none`: no control.
- `:I`, `:Ih`: rogue all infective, or only infective-high, vines.
- `:ImY`: rogue infective-moderate young vines.
- `:ImNS`, `:ImNSEW`: rogue each infective-moderate vine and test its two neighbours along the row (or all four neighbours), roguing those found exposed-detectable.
- `:ImNS2EW`: also test the second neighbours along the row.

```
strategies = (:none, :I, :Ih, :ImY, :ImNS, :ImNSEW, :ImNS2EW)
source = Dict(:none => 316, :I => -147, :Ih => 284, :ImY => 314, :ImNS => 374,
              :ImNSEW => 374, :ImNS2EW => 309)
enpv(strategy; runs=20, kw...) = [grapevine_npv(solve(problem(:grapevine_leafroll;
strategy,
                                                              seed=s, kw...)))
                                  for s in 1:runs] ./ 1000
results = Dict(st => enpv(st) for st in strategies)
[(strategy=st, model=round(mean(results[st])), sd=round(std(results[st]); digits=1),
  source=source[st]) for st in strategies]
```

```
7-element Vector{@NamedTuple{strategy::Symbol, model::Float64, sd::Float64,
source::Int64}}:
 (strategy = :none, model = 340.0, sd = 3.2, source = 316)
 (strategy = :I, model = 382.0, sd = 3.4, source = -147)
 (strategy = :Ih, model = 330.0, sd = 3.8, source = 284)
 (strategy = :ImY, model = 348.0, sd = 3.0, source = 314)
 (strategy = :ImNS, model = 382.0, sd = 4.8, source = 374)
 (strategy = :ImNSEW, model = 382.0, sd = 5.0, source = 374)
 (strategy = :ImNS2EW, model = 382.0, sd = 4.3, source = 309)
```

Testing the neighbours along the row comes out as in the source: about $382k against $374k, a gain over no control. Roguing infective-high vines alone loses value, as in the source. Several results differ:

- With the slower epidemic, no control is worth more than in the source.

- Testing the second neighbours along the row is worth as much as testing the first, where the source finds it costs more than it saves.
- Roguing young symptomatic vines gains a little, where the source finds it loses a little.
- Roguing every infective vine, which the source finds ruinous at −$147k, is as good here as the spatial strategies.

6 Choices the source leaves open

The source does not say when a symptomatic vine is found and rogued. By default it is rogued after it has been infective for one month, so it infects its row neighbours first (`delay = 1`). With `delay = 0` it is rogued as soon as it shows symptoms, and the spatial strategies look much better than they do in the source:

```
[(strategy=st, delay_0=round(mean(enpv(st; runs=10, delay=0))),
  delay_1=round(mean(results[st]))) for st in (:none, :I, :ImNS)]
```

```
3-element Vector{@NamedTuple{strategy::Symbol, delay_0::Float64, delay_1::Float64}}:
 (strategy = :none, delay_0 = 339.0, delay_1 = 340.0)
 (strategy = :I, delay_0 = 457.0, delay_1 = 382.0)
 (strategy = :ImNS, delay_0 = 443.0, delay_1 = 382.0)
```

The first infections are “exposed” in the source, which does not say whether they can be detected. Starting them infective (`initial = :Im`) brings the half-life and the no-control value close to the source’s (188 months and $316k). But testing neighbours then gains much less than in the source:

```
hl_im = [grapevine_half_life(solve(problem(:grapevine_leafroll; initial=:Im,
seed=s)))
         for s in 1:20]
(half_life=mean(hl_im),
 none=round(mean(enpv(:none; runs=10, initial=:Im))),
 ImNS=round(mean(enpv(:ImNS; runs=10, initial=:Im))))
```

```
(half_life = 186.7, none = 307.0, ImNS = 321.0)
```

No reading tried reproduces every row of the source’s table. Most of all, none makes roguing every infective vine as costly as it is in the source.

7 Deviations from the source

```
foreach(d -> println("- ", d), model_info(:grapevine_leafroll).deviations)
```

```
- The journal article was not read (paywalled); the model follows Chapter 2 of the
first author's dissertation (Atallah 2014), whose abstract-level results match the
article's.
- The undetectable period is written in the source as a cumulative probability of
time in the state; by default it is drawn as a triangular duration (4, 12, 18
months, rounded up) when a vine is infected, and undetectable = :cdf applies the
source's formula as a monthly probability.
- The initial infections start exposed-undetectable; the source says only 'exposed'.
With this start the epidemic is slower than the source's (half-life about 234 months
against 188); starting them infective (initial = :Im) matches the source's half-life
and no-control value but not its spatial strategies.
- The source does not say when a symptomatic vine is found; by default it is rogued
after it has been infective for a month (delay = 1), so it infects its neighbours
first. This reproduces the source's spatial strategies (about $380k against $374k).
No reading tried reproduces the source's large losses from roguing every infective
vine (strategy I, about -$147k in the source).
- The order within a month is not stated: revenue is earned on the state at the
start of the month, then the disease steps, then management acts on the new state
and its costs are charged that month.
- A replant is not rogued before it is 36 months old (the source's constraint that a
replant is never rogued before it bears fruit); protect_replants = false lifts this.
- Each neighbour of a triggering vine is tested at most once a month; vines already
infective are not tested.
- Vines are productive from 36 months; the source's revenue table leaves 37 to 47
months undefined. This reproduces its no-disease value.
- The two-vineyard extension of the source's Chapter 3 is not implemented: its
long-distance dispersal equations are inconsistent.
```


Atallah, Shady S. 2014. "Plant-Level, Spatial, Bioeconomic Models of Plant Disease Diffusion and Control: Grapevine Leafroll Disease." PhD thesis, Cornell University. https://hdl.handle.net/1813/38920.

Atallah, Shady S., Miguel I. Gómez, Jon M. Conrad, and Jan P. Nyrop. 2015. "A Plant-Level, Spatial, Bioeconomic Model of Plant Disease Diffusion and Control: Grapevine Leafroll Disease." *American Journal of Agricultural Economics* 97 (1): 199–218. https://doi.org/10.1093/ajae/aau032.